

23-S1-Q1

Q: TSP 12 city $\langle a_1 a_2 \dots a_{12} \rangle$

recombination $\rightarrow$ hyper-recomb

① PM X : shade cells

② cycle $\sim$

③ Edge. $\sim \rightarrow$ Node 1

Node 7

greedy approach : lowest cost

$6 \rightarrow 2$ dij % : modulo

$$d_{ij} = ((a_i + a_j) \% 5) + 1$$

ca) two offspring

Parent 1 5 12 7 1 10 3 8 6 11 4 9 2

Parent 2 7 8 9 10 11 12 1 2 3 4 5 6

Solution ① PMX off spring 1

Parent 1 5 12 7 1 10 3 8 6 11 4 9 2

Parent 2 7 8 9 10 11 12 1 2 3 4 5 6

Son — 12 — 1 10 3 8 6 11 — — —

Parent 1 5 12 7 1 10 3 8 6 11 4 9 2

Parent 2 7 8 9 10 11 12 1 2 3 4 5 6

Son — 12 — 1 10 3 8 6 11 — — 2

Parent 1 5 12 7 1 10 3 8 6 11 4 9 2

Parent 2 7 8 9 10 11 12 1 2 3 4 5 6

Son 7 12 9 1 10 3 8 6 11 4 5 2

② PMX off spring 2

Parent 1 5 12 7 1 10 3 8 6 11 4 9 2

Parent 2 7 8 9 10 11 12 1 2 3 4 5 6

Son — 8 — 10 11 12 1 2 3 — — —

Parent 1 5 12 7 ~~1~~ ~~10~~ ~~3~~ ~~8~~ ~~6~~ ~~11~~ 4 9 2

Parent 2 7 8 9 ~~10~~ ~~11~~ ~~12~~ ~~1~~ ~~2~~ ~~3~~ 4 5 6

Son — 8 — 10 11 12 1 2 3 — — 6

Parent 1 5 12 7 ~~1~~ ~~10~~ ~~3~~ ~~8~~ ~~6~~ ~~11~~ 4 9 2

Parent 2 7 8 9 ~~10~~ ~~11~~ ~~12~~ ~~1~~ ~~2~~ ~~3~~ 4 5 6

Son 5 8 7 10 11 12 1 2 3 4 9 6

③ After PMX

offspring 1 7 12 9 1 10 3 8 6 11 4 5 2

offspring 2 5 8 7 10 11 12 1 2 3 4 9 6

④ Cycle recombination: cycle 1

Parent 1 5 12 7 1 10 3 8 6 11 4 9 2

Parent 2 7 8 9 10 11 12 1 2 3 4 5 6

⑤ Cycle 2

Parent 1 5 12 7 1 10 3 8 6 11 4 9 2
 Parent 2 7 8 9 10 11 12 1 2 3 4 5 6

⑥ Cycle 3

Parent 1 5 12 7 1 10 3 8 6 11 4 9 2
 Parent 2 7 8 9 10 11 12 1 2 3 4 5 6

⑦ Cycle 4

Parent 1 5 12 7 1 10 3 8 6 11 4 9 2
 Parent 2 7 8 9 10 11 12 1 2 3 4 5 6

⑧ fix 1, 3 chang 2, 4

offspring 3 5 8 7 10 11 12 1 6 3 4 9 2
 offspring 4 7 12 9 1 10 3 8 2 11 4 5 6

⑨ After Circle

offspring 3 5 8 7 10 11 12 1 6 3 4 9 2
 offspring 4 7 12 9 1 10 3 8 2 11 4 5 6

⑩ Edge : start with 1 & 7

⑪ offspring

Parent 1 5 12 7 1 10 3 8 6 11 4 9 2

Parent 2 7 8 9 10 11 12 1 2 3 4 5 6

1	7	10	12	2
2	9	5	1	3
3	10	8	2	4
4	11	9	3	5
5	12	2	4	6
6	8	11	5	7
7	12	1	8	6
8	3	6	7	9
9	4	2	8	10
10	1	3	9	11
11	6	4	10	12
12	5	7	11	1

Choices	Element Selected	Reason	Partial Result
1	1	offspring start with 1	1
7 10 12 2	10	smallest d	1 10
3 9 11	11	Smallest d	1 10 11
6 4 12	4	Smallest d	1 10 11 4
9 3 5	3	Smallest d	1 10 11 4 3
8 2	2	Smallest d	1 10 11 4 3 2
9 5	9	Smallest d	1 10 11 4 3 2 9
8	8	only one	1 10 11 4 3 2 9 8
6, 7	7	Smallest d	1 10 11 4 3 2 9 8 7
12, 6	6	Smallest d	1 10 11 4 3 2 9 8 7 6
5	5	only one	1 10 11 4 3 2 9 8 7 6 5
12	12	only one	1 10 11 4 3 2 9 8 7 6 5 12

$$((i+j) \bmod 5) + 1$$

x	11
7	18 → 3 → 4
10	21 → 1 → 2 ✓
12	23 → 3 → 4
2	13 → 3 → 4

x	10
3	13 → 3 → 4
9	19 → 4 → 5
11	21 → 1 → 2 ✓

x	11
6	17 → 2 → 3
4	15 → 0 → 1 ✓
12	23 → 3 → 4

	4
9	13 → 3 → 4
3	7 → 2 → 3 ✓
5	9 → 4 → 5

	3
8	11 → 1 → 2
2	5 → 0 → 1 ✓

	2
9	11 → 1 → 2 ✓
5	7 → 2 → 3

	9
8	✓

	8
6	14 → 4 → 5
7	15 → 0 → 1 ✓

	7
12	19 → 4 → 5
6	13 → 3 → 4 ✓

	6
5	✓

	5
12	✓

offspring 5 1 10 9 2 3 8 7 6 5 12 11 4

⑫ offspring : similar step!
skip

1	7	10	12	2
2	9	5	1	3
3	10	8	2	4
4	11	9	3	5
5	12	2	4	6
6	8	11	5	7
7	12	1	8	6
8	3	6	7	9
9	4	2	8	10
10	1	3	9	11
11	6	4	10	12
12	5	7	11	1

Choices	Element Selected	Reason	Partial Result
7	7	offspring 6 start with 7	7
12 1 8 6	8	mini d	78
3 6 9	3	~	783
10 2 4	2	~	7832
9 5 1	9	~	78329
4 10	4	~	783294
11 5	11	~	78329411
6 10 12	10	~	7832941110
1	1	only	78329411101
12	12	only	7832941110112
5 7	5	~	78329411101125
6	6	only	783294111011256

	7
12	5
1	4
8	1
6	4

	8
3	2
6	5
9	3

	3
10	4
2	1
4	3

	2
9	2
5	3
1	4

	9
4	4
10	5

	4
11	1
5	5

	11
6	3
10	2
12	4

	12
5	3
7	5

⑬ cost computations

offspring 1 7 12 9 1 10 3 8 6 11 4 5 2

offspring 2 5 8 7 10 11 12 1 2 3 4 9 6

offspring 3 5 8 7 10 11 12 1 6 3 4 9 2

offspring 4 7 12 9 1 10 3 8 2 11 4 5 6

offspring 5 1 10 11 4 3 2 9 8 7 6 5 12

offspring 6 7 8 3 2 9 4 11 10 1 12 5 6

⑭ greedy compute for each offspring

offspring 1 7 12 9 1 10 3 8 6 11 4 5 2

dis_{i,j} 5 2 1 2 4 2 5 3 5 5 3

Sum 37

offspring 2 5 8 7 10 11 12 1 2 3 4 9 6

dis_{i,j} 4 1 3 2 4 4 4 1 3 4 1

Sum 31

offspring 3
 5 8 7 10 11 12 1 6 3 4 9 2
 4 1 3 2 4 4 3 5 3 4 2
 sum 35

offspring 4
 7 12 9 1 10 3 8 2 11 4 5 6
 5 2 1 2 4 2 1 4 1 5 2
 sum 29

offspring 5
 1 10 11 4 3 2 9 8 7 6 5 12
 2 2 1 3 1 2 3 1 4 2 3
 sum 24

offspring 6
 7 8 3 2 9 4 11 10 1 12 5 6
 1 2 1 2 4 1 2 2 4 3 2
 sum 24

offspring	Sum of dij	
1	37	
2	31	
3	35	
4	29	
5	24	✓
6	24	✓

So we choose
 offspring 5 and 6

(b) Pros & Cons

Pros ① Greater Diversity of Solution

② Potentially Faster Improvement

③ Combine three methods pros

(1) PMX often preserves relative order of substrings nicely

(2) Cycle crossover systematically passes certain "alleles" in cycles

(3) Edge ~ preserve adjacency information

Cons

① increased computational cost per Pair

② Risk of Overfitting to the crossover stage

③ complexity of Implement and tune